Lesson 56: Natural Policy Gradient

AI/ML Mathematics

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1 Motivation and Overview

Consider training a policy π_θ for a locomotion task via the vanilla policy gradient $\theta_{k+1} = \theta_k + \alpha \nabla_\theta J(\theta_k)$. In practice, a small step in parameter space $\|\Delta\theta\| = \epsilon$ can produce an enormous change in the policy distribution when the Fisher information matrix has large eigenvalues, or almost no change when it has small eigenvalues. The steepest-ascent direction in Euclidean parameter space is therefore poorly matched to the geometry of the policy manifold: the same step size α that works for one parameterization may be catastrophically large or wastefully small for another. What we need is a gradient that is *covariant* – that descends in the same direction regardless of how we parameterize the policy. The natural gradient of ichi Amari [1998], adapted to policy optimization by Kakade [2002], provides exactly this: by preconditioning the gradient with the inverse Fisher information matrix, we obtain the steepest ascent direction in the space of probability distributions rather than the space of parameters.

RL/ML Connection

- **Natural actor-critic** [Peters and Schaal, 2008]: Combines the natural policy gradient with compatible function approximation, solving a least-squares critic problem to obtain the natural gradient direction in closed form.
- **TRPO** [Schulman et al., 2015]: Approximates the natural policy gradient via a trust-region constraint on the KL divergence, using conjugate gradients to solve the Fisher-vector product.
- **K-FAC for policy optimization**: Approximates the Fisher information matrix using Kronecker-factored representations for efficient natural gradient computation in deep networks.
- **PPO** [Schulman et al., 2017]: Can be viewed as an approximate natural gradient method where the clipped surrogate implicitly limits the KL step size.

Prerequisites. The reader must know the policy gradient theorem in both the log-derivative and expectation form (Lesson 54, Theorem 5.1), the baseline invariance proposition (Lesson 54, Proposition 7.1), the compatible function approximation theorem (Lesson 54, Theorem 8.2), and the advantage function (Lesson 55, Definition 4.3). From linear algebra, we need positive semi-definite matrices, eigenvalue decompositions, and the Moore-Penrose pseudoinverse (Lesson 7). Familiarity with the KL divergence and its second-order expansion (Lesson 12) is assumed.

Learning objectives.

- **Explain** why the vanilla policy gradient is not covariant under reparameterization and why this is problematic.
- **Define** the Fisher information matrix for a parameterized policy and derive its key properties (positive semi-definiteness, relation to KL divergence Hessian).
- **Derive** the natural policy gradient as the solution to the steepest-ascent problem under a KL divergence constraint.
- **Prove** that the natural policy gradient is covariant under smooth invertible reparameterizations.
- **State** and prove the connection between compatible function approximation and the natural gradient.

- **Compute** the Fisher information matrix and natural gradient update for softmax and Gaussian policy parameterizations.

2 References

- ichi Amari [1998] – The original natural gradient paper; defines the Fisher metric and derives the natural gradient update for general statistical models.
- Kakade [2002] – Adapts the natural gradient to policy optimization; introduces the compatible function approximation connection.
- Kakade [2003, Chapter 2] – Extended treatment of natural policy gradients with sample complexity analysis.
- Peters and Schaal [2008] – Natural actor-critic algorithms; practical implementations using compatible features.
- Bhatnagar et al. [2009] – Convergence analysis of natural actor-critic algorithms.
- ichi Amari [2016] – Comprehensive reference on information geometry; Chapters 1–3 cover the Fisher metric and its properties.

3 The Reparameterization Problem

We begin by demonstrating that the vanilla policy gradient is not invariant under reparameterization, motivating the need for a geometry-aware gradient.

Definition 3.1 (Reparameterization)**.** Let π_θ be a parameterized policy with $\theta \in \Theta \subseteq \mathbb{R}^d$, and let $\varphi : \Phi \to \Theta$ be a smooth invertible map (a diffeomorphism). The *reparameterized* policy is $\pi_{\varphi(\eta)} = \tilde{\pi}_\eta$, parameterized by $\eta \in \Phi$.

Proposition 3.2 (Non-Covariance of the Vanilla Gradient)**.** *Under the reparameterization $\theta = \varphi(\eta)$, the gradient of the objective $\tilde{J}(\eta) = J(\varphi(\eta))$ transforms as*

$$\nabla_\eta \tilde{J}(\eta) = \left(\frac{\partial \varphi}{\partial \eta}\right)^\top \nabla_\theta J(\theta).$$

Consequently, the vanilla gradient ascent direction depends on the parameterization: the update $\Delta\eta = \alpha \nabla_\eta \tilde{J}$ corresponds to $\Delta\theta = \alpha\, (\partial\varphi/\partial\eta)^\top \nabla_\theta J$ in the original space, which generally differs from $\alpha \nabla_\theta J$.

Proof. By the chain rule:

$$\nabla_\eta \tilde{J}(\eta) = \nabla_\eta J(\varphi(\eta)) = \left(\frac{\partial \varphi}{\partial \eta}\right)^\top \nabla_\theta J(\theta),$$

where $\partial\varphi/\partial\eta \in \mathbb{R}^{d\times d}$ is the Jacobian matrix. The transformed gradient direction is $(\partial\varphi/\partial\eta)^\top \nabla_\theta J$, which differs from $\nabla_\theta J$ unless the Jacobian is the identity. □

Example 3.3 (Scaling Reparameterization)**.** Consider a one-dimensional parameter $\theta \in \mathbb{R}$ and the reparameterization $\theta = c\,\eta$ for some constant $c > 0$. Then $\nabla_\eta \tilde{J} = c\,\nabla_\theta J$. A gradient step $\eta \leftarrow \eta + \alpha \nabla_\eta \tilde{J}$ corresponds to $\theta \leftarrow \theta + \alpha c^2 \nabla_\theta J$ – the effective step size is scaled by c^2. If $c = 100$, the update in the original space is 10,000 times larger. This simple example shows how the choice of parameterization can drastically affect optimization behavior.

RL/ML Connection

In deep RL, reparameterizations arise naturally from different network architectures, initializations, and activation functions. The non-covariance of vanilla gradients means optimization can be highly sensitive to these choices, motivating parameterization-invariant updates.

4 The Fisher Information Matrix

The Fisher information matrix provides the natural Riemannian metric on the space of parameterized probability distributions.

Definition 4.1 (Fisher Information Matrix)**.** For a parameterized policy $\pi_\theta(a \mid s)$, the *Fisher information matrix* at state s is

$$F_s(\theta) = \mathbb{E}_{a\sim\pi_\theta(\cdot|s)}\Big[\nabla_\theta \log \pi_\theta(a \mid s)\, \nabla_\theta \log \pi_\theta(a \mid s)^\top\Big] \in \mathbb{R}^{d\times d}.$$

The *average Fisher information matrix* under the on-policy state distribution d^{π_θ} is

$$F(\theta) = \mathbb{E}_{s\sim d^{\pi_\theta}}\big[F_s(\theta)\big] = \mathbb{E}_{s\sim d^{\pi_\theta},\, a\sim\pi_\theta}\Big[\nabla_\theta \log \pi_\theta(a \mid s)\, \nabla_\theta \log \pi_\theta(a \mid s)^\top\Big].$$

Proposition 4.2 (Properties of the Fisher Information Matrix)**.** *The Fisher information matrix $F(\theta)$ satisfies:*

(i) ***Positive semi-definiteness:*** *$F(\theta) \succeq 0$ for all θ.*

(ii) ***Covariance interpretation:*** *$F_s(\theta) = \mathrm{Cov}_{a\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a \mid s)]$.*

(iii) ***Hessian of negative log-likelihood:*** *$F_s(\theta) = -\mathbb{E}_{a\sim\pi_\theta}[\nabla^2_\theta \log \pi_\theta(a \mid s)]$, provided the interchange of differentiation and expectation is valid.*

Proof. (i) For any $v \in \mathbb{R}^d$:

$$v^\top F_s(\theta)\, v = \mathbb{E}_a\Big[(v^\top \nabla_\theta \log \pi_\theta(a \mid s))^2\Big] \geq 0.$$

Averaging over s preserves the inequality, so $v^\top F(\theta) v \geq 0$.

(ii) By Lemma 4.2 of Lesson 54, $\mathbb{E}_a[\nabla_\theta \log \pi_\theta(a \mid s)] = 0$. Therefore:

$$\begin{aligned} F_s(\theta) &= \mathbb{E}_a\Big[\nabla_\theta \log \pi_\theta\, (\nabla_\theta \log \pi_\theta)^\top\Big] - \underbrace{\mathbb{E}_a[\nabla_\theta \log \pi_\theta]}_{=0}\, \underbrace{\mathbb{E}_a[\nabla_\theta \log \pi_\theta]^\top}_{=0} \\ &= \mathrm{Cov}_a[\nabla_\theta \log \pi_\theta(a \mid s)]. \end{aligned}$$

(iii) Using $\nabla^2_\theta \log \pi_\theta = \nabla^2_\theta \pi_\theta / \pi_\theta - (\nabla_\theta \log \pi_\theta)(\nabla_\theta \log \pi_\theta)^\top$ and noting $\sum_a \nabla^2_\theta \pi_\theta = \nabla^2_\theta 1 = 0$:

$$\mathbb{E}_a[\nabla^2_\theta \log \pi_\theta] = \underbrace{\sum_a \nabla^2_\theta \pi_\theta}_{=0} - F_s(\theta) = -F_s(\theta). \qquad \square$$

Theorem 4.3 (Fisher Information as KL Hessian)**.** *The Fisher information matrix equals the Hessian of the KL divergence between neighboring policies:*

$$F_s(\theta) = \nabla^2_{\theta'}\, \mathrm{KL}\big(\pi_\theta(\cdot \mid s) \,\|\, \pi_{\theta'}(\cdot \mid s)\big)\Big|_{\theta'=\theta}.$$

In particular, the KL divergence admits the local quadratic approximation

$$\mathrm{KL}\big(\pi_\theta(\cdot \mid s) \,\|\, \pi_{\theta+\Delta\theta}(\cdot \mid s)\big) \approx \frac{1}{2}\, \Delta\theta^\top F_s(\theta)\, \Delta\theta.$$

Proof. Differentiating $\mathrm{KL}(\pi_\theta \| \pi_{\theta'}) = \sum_a \pi_\theta(a \mid s) \log(\pi_\theta / \pi_{\theta'})$ with respect to θ': $\nabla_{\theta'} \mathrm{KL} = -\sum_a \pi_\theta \nabla_{\theta'} \log \pi_{\theta'}$ and $\nabla^2_{\theta'} \mathrm{KL} = -\sum_a \pi_\theta \nabla^2_{\theta'} \log \pi_{\theta'}$. At $\theta' = \theta$: $\nabla^2_{\theta'} \mathrm{KL}\big|_{\theta'=\theta} = -\mathbb{E}_\pi[\nabla^2_\theta \log \pi_\theta] = F_s(\theta)$ by Proposition 4.2(iii). Since $\mathrm{KL}(\pi_\theta \| \pi_\theta) = 0$ and the first derivative vanishes at $\theta' = \theta$ (zero-mean score), the Taylor expansion gives $\mathrm{KL}(\pi_\theta \| \pi_{\theta+\Delta\theta}) \approx \frac{1}{2} \Delta\theta^\top F_s(\theta) \Delta\theta$. □

Fisher Information as KL Curvature

The Fisher information matrix $F(\theta)$ captures the local curvature of the KL divergence between the current policy and nearby policies. A direction v with large $v^\top F(\theta) v$ corresponds to a parameter perturbation that changes the policy distribution rapidly; a direction with small curvature changes the distribution slowly. The Fisher metric thus provides a notion of "distance in distribution space" that is more meaningful for optimization than Euclidean distance in parameter space.

Example 4.4 (Fisher Information for Softmax Policy)**.** For the softmax policy $\pi_\theta(a \mid s) = \exp(\theta^\top \phi(s,a)) / \sum_{a'} \exp(\theta^\top \phi(s,a'))$ with score function $\nabla_\theta \log \pi_\theta(a \mid s) = \phi(s,a) - \bar{\phi}_\theta(s)$ where $\bar{\phi}_\theta(s) = \mathbb{E}_{a' \sim \pi_\theta}[\phi(s,a')]$ (Lesson 54, Example 4.5), the Fisher information at state s is:

$$\begin{aligned} F_s(\theta) &= \mathbb{E}_{a \sim \pi_\theta}\Big[(\phi(s,a) - \bar{\phi}_\theta(s))(\phi(s,a) - \bar{\phi}_\theta(s))^\top\Big] \\ &= \mathrm{Cov}_{a \sim \pi_\theta}\big[\phi(s,a)\big]. \end{aligned}$$

The Fisher information is the covariance matrix of the feature vectors under the current policy.

Example 4.5 (Fisher Information for Gaussian Policy)**.** For the Gaussian policy $\pi_\theta(a \mid s) = \mathcal{N}(a;\, \theta^\top \phi(s),\, \sigma^2 I)$ with fixed variance σ^2 and score function $\nabla_\theta \log \pi_\theta(a \mid s) = \frac{(a - \theta^\top \phi(s))}{\sigma^2} \phi(s)$ (scalar action case, from Lesson 54, Example 4.6), the Fisher information at state s is:

$$F_s(\theta) = \mathbb{E}_{a \sim \pi_\theta}\left[\frac{(a - \theta^\top \phi(s))^2}{\sigma^4} \phi(s)\, \phi(s)^\top\right] = \frac{1}{\sigma^2} \phi(s)\, \phi(s)^\top,$$

since $\mathbb{E}[(a-\mu)^2] = \sigma^2$. The average Fisher information is $F(\theta) = \frac{1}{\sigma^2} \mathbb{E}_{s \sim d^\pi}[\phi(s)\, \phi(s)^\top]$, which is $1/\sigma^2$ times the feature covariance matrix under the state distribution.

5 The Natural Gradient

We now derive the natural gradient as the steepest-ascent direction in the Riemannian manifold defined by the Fisher metric.

Definition 5.1 (Steepest Ascent under a Metric)**.** Given a positive definite matrix $G(\theta) \succ 0$ defining a local inner product $\langle u, v \rangle_G = u^\top G\, v$, the *steepest ascent direction* of $J(\theta)$ under this metric is

$$\Delta\theta^* = \arg\max_{\Delta\theta}\ \nabla_\theta J(\theta)^\top \Delta\theta \quad \text{subject to} \quad \Delta\theta^\top G(\theta)\, \Delta\theta \le \epsilon.$$

Theorem 5.2 (Natural Gradient)**.** *The steepest ascent direction under the Fisher metric $G(\theta) = F(\theta)$ is*

$$\Delta\theta^* \propto F(\theta)^{-1} \nabla_\theta J(\theta).$$

The vector $\tilde{\nabla}_\theta J(\theta) = F(\theta)^{-1} \nabla_\theta J(\theta)$ is called the natural policy gradient.

Proof. The constrained optimization problem in Definition 5.1 has the Lagrangian

$$\mathcal{L}(\Delta\theta, \lambda) = \nabla_\theta J^\top \Delta\theta - \lambda\, (\Delta\theta^\top F\, \Delta\theta - \epsilon).$$

Setting $\nabla_{\Delta\theta} \mathcal{L} = 0$:

$$\nabla_\theta J - 2\lambda\, F\, \Delta\theta = 0 \quad \Longrightarrow \quad \Delta\theta = \frac{1}{2\lambda} F^{-1} \nabla_\theta J.$$

The multiplier $\lambda > 0$ is determined by the constraint $\Delta\theta^\top F\,\Delta\theta = \epsilon$, which fixes the step length but not the direction. Therefore $\Delta\theta^* \propto F^{-1}\,\nabla_\theta J$. □

Remark 5.3. By Theorem 4.3, the constraint $\Delta\theta^\top F(\theta)\,\Delta\theta \le \epsilon$ is equivalent (to second order) to the KL divergence constraint

$$\mathbb{E}_{s\sim d^\pi}\left[\mathrm{KL}\big(\pi_\theta(\cdot \mid s) \,\|\, \pi_{\theta+\Delta\theta}(\cdot \mid s)\big)\right] \le \frac{\epsilon}{2}.$$

The natural gradient is therefore the steepest ascent direction under the constraint that the policy distribution changes by at most a fixed amount, as measured by KL divergence. This is the starting point for trust-region methods (Lesson 58).

Natural Policy Gradient Update

The natural policy gradient update is

$$\theta_{k+1} = \theta_k + \alpha\, F(\theta_k)^{-1}\,\nabla_\theta J(\theta_k),$$

where $F(\theta_k)$ is the Fisher information matrix and $\alpha > 0$ is the step size. This update takes the steepest ascent step in distribution space (as measured by KL divergence) rather than in Euclidean parameter space.

Invertibility of the Fisher Matrix

The Fisher information matrix $F(\theta)$ is positive semi-definite but may be singular (e.g., when the parameterization is redundant). In practice, a damping term is added: $\tilde\nabla_\theta J = (F(\theta) + \lambda I)^{-1}\nabla_\theta J$ for small $\lambda > 0$. This ensures invertibility and is equivalent to adding an L^2 regularization term in the steepest-ascent problem. Alternatively, one can use the Moore-Penrose pseudoinverse $F^\dagger$.

6 Covariance of the Natural Gradient

The key property that distinguishes the natural gradient from the vanilla gradient is its invariance under reparameterization.

Theorem 6.1 (Covariance under Reparameterization)**.** *Let $\theta = \varphi(\eta)$ be a smooth invertible reparameterization with Jacobian $J_\varphi = \partial\varphi/\partial\eta$. Then the natural gradient is covariant:*

$$\tilde\nabla_\eta \tilde J(\eta) = J_\varphi^{-1}\,\tilde\nabla_\theta J(\theta).$$

In particular, the natural gradient update applied in the η parameterization maps to the same natural gradient update in the θ parameterization (up to a reparameterization of the step).

Proof. **Step 1: Gradient transformation.** By the chain rule, $\nabla_\eta \tilde J = J_\varphi^\top\,\nabla_\theta J$.

Step 2: Fisher transformation. The score function transforms as $\nabla_\eta \log\tilde\pi_\eta = J_\varphi^\top\,\nabla_\theta \log\pi_\theta$, so:

$$\begin{aligned}\tilde F(\eta) &= \mathbb{E}\Big[\nabla_\eta \log\tilde\pi_\eta\,(\nabla_\eta \log\tilde\pi_\eta)^\top\Big] \\ &= J_\varphi^\top\,\mathbb{E}\Big[\nabla_\theta \log\pi_\theta\,(\nabla_\theta \log\pi_\theta)^\top\Big]\,J_\varphi = J_\varphi^\top\,F(\theta)\,J_\varphi.\end{aligned}$$

Step 3: Natural gradient in η coordinates.

$$\begin{aligned}\tilde\nabla_\eta \tilde J &= \tilde F(\eta)^{-1}\,\nabla_\eta \tilde J = (J_\varphi^\top F J_\varphi)^{-1}\,J_\varphi^\top \nabla_\theta J \\ &= J_\varphi^{-1}\,F^{-1}\,(J_\varphi^\top)^{-1}\,J_\varphi^\top\,\nabla_\theta J = J_\varphi^{-1}\,F^{-1}\,\nabla_\theta J = J_\varphi^{-1}\,\tilde\nabla_\theta J.\end{aligned}$$

The update $\Delta\eta = \alpha\,\tilde{\nabla}_\eta \tilde{J}$ maps to $\Delta\theta = J_\varphi\,\Delta\eta = \alpha\,\tilde{\nabla}_\theta J$ in the original coordinates, confirming covariance. □

Example 6.2 (Covariance Verification for Scaling)**.** For $\theta = c\,\eta$ (Example 3.3): $J_\varphi = c$, $\tilde{F}(\eta) = c^2 F(\theta)$, and $\tilde{\nabla}_\eta \tilde{J} = (c^2 F)^{-1}(c\,\nabla_\theta J) = \frac{1}{c}\tilde{\nabla}_\theta J$. The update $\Delta\theta = c\,\Delta\eta = \alpha\,\tilde{\nabla}_\theta J$ is identical to the natural gradient in the original parameterization, unlike vanilla which scales by c^2.

7 Compatible Function Approximation and the Natural Gradient

A remarkable connection exists between the compatible function approximation theorem (Lesson 54, Theorem 8.2) and the natural gradient: the weight vector of the optimal compatible critic is exactly the natural gradient direction.

Theorem 7.1 (Natural Gradient via Compatible Features)**.** *Let $f_w(s,a) = w^\top \nabla_\theta \log \pi_\theta(a \mid s)$ be the compatible function approximator, and let w^* minimize the mean squared error*

$$\varepsilon(w) = \mathbb{E}_{s\sim d^\pi,\, a\sim\pi_\theta}\Big[\big(f_w(s,a) - A^\pi(s,a)\big)^2\Big].$$

Then $w^* = \tilde{\nabla}_\theta J(\theta) = F(\theta)^{-1}\,\nabla_\theta J(\theta)$, the natural policy gradient.

Proof. **Step 1: Optimality condition.** Setting $\nabla_w \varepsilon(w) = 0$:

$$\mathbb{E}_{s,a}\Big[\nabla_\theta \log \pi_\theta(a \mid s)\,\big(w^\top \nabla_\theta \log \pi_\theta(a \mid s) - A^\pi(s,a)\big)\Big] = 0.$$

Step 2: Expanding.

$$\begin{aligned}\mathbb{E}_{s,a}&\Big[\nabla_\theta \log \pi_\theta\,(\nabla_\theta \log \pi_\theta)^\top\Big]\,w \\ &= \mathbb{E}_{s,a}[\nabla_\theta \log \pi_\theta\, A^\pi(s,a)]\,.\end{aligned}$$

The left-hand matrix is $F(\theta)$. By the policy gradient theorem with the advantage form (Lesson 55, Proposition 4.5), the right-hand side equals $\nabla_\theta J(\theta)$.

Step 3: Solving.

$$F(\theta)\,w^* = \nabla_\theta J(\theta) \quad\Longrightarrow\quad w^* = F(\theta)^{-1}\,\nabla_\theta J(\theta) = \tilde{\nabla}_\theta J(\theta).$$ □

Compatible Features Yield the Natural Gradient

The natural policy gradient $\tilde{\nabla}_\theta J(\theta) = F(\theta)^{-1}\,\nabla_\theta J(\theta)$ equals the weight vector w^* that solves the compatible function approximation least-squares problem. This provides a practical algorithm: instead of explicitly computing and inverting $F(\theta)$, one can solve a regression problem to obtain the natural gradient direction.

RL/ML Connection

The natural actor-critic of Peters and Schaal [2008] exploits Theorem 7.1: the critic learns w via TD on compatible features $f_w(s,a) = w^\top \nabla_\theta \log \pi_\theta(a \mid s)$; at convergence $w \approx w^*$, and the actor updates $\theta \leftarrow \theta + \alpha\,w$. This avoids explicit computation and inversion of $F(\theta)$.

Example 7.2 (Natural Gradient for Gaussian Policy)**.** For the Gaussian policy with $\pi_\theta(a \mid s) = \mathcal{N}(a;\, \theta^\top \phi(s),\, \sigma^2)$ (scalar action), the Fisher information is $F = \frac{1}{\sigma^2}\, \mathbb{E}_s[\phi(s)\, \phi(s)^\top] = \frac{1}{\sigma^2}\, \Phi$ where Φ is the feature covariance (Example 4.5). The natural gradient is:

$$\tilde{\nabla}_\theta J = F^{-1}\, \nabla_\theta J = \sigma^2\, \Phi^{-1}\, \nabla_\theta J.$$

This removes the $1/\sigma^2$ scaling present in the vanilla gradient, making the natural gradient independent of the noise level σ^2. As training proceeds and σ^2 decreases, the vanilla gradient shrinks while the natural gradient remains well-scaled.

8 The Natural Policy Gradient Algorithm

We now present the complete natural policy gradient algorithm and analyze its computational requirements.

Definition 8.1 (Natural Policy Gradient Algorithm)**.** The natural policy gradient algorithm iterates:

1. Collect trajectory data $\{(s_t, a_t, r_t)\}$ under π_{θ_k}.
2. Estimate the policy gradient $\hat{g}_k \approx \nabla_\theta J(\theta_k)$ using the REINFORCE or advantage estimator.
3. Estimate the Fisher information matrix $\hat{F}_k \approx F(\theta_k)$ from the same trajectory data.
4. Compute the natural gradient $\hat{w}_k = (\hat{F}_k + \lambda I)^{-1}\, \hat{g}_k$.
5. Update: $\theta_{k+1} = \theta_k + \alpha_k\, \hat{w}_k$.

Proposition 8.2 (Sample Fisher Estimate)**.** *Given N state-action pairs $\{(s_i, a_i)\}_{i=1}^N$ sampled from the on-policy distribution, $\hat{F} = \frac{1}{N} \sum_{i=1}^N \nabla_\theta \log \pi_\theta(a_i \mid s_i)\, \nabla_\theta \log \pi_\theta(a_i \mid s_i)^\top$ is an unbiased estimate of $F(\theta)$. Forming $\hat{F}$ costs $O(Nd^2)$ and inverting it costs $O(d^3)$. For large d, the conjugate gradient method solves $\hat{F}\, w = \hat{g}$ using only Fisher-vector products, each costing $O(Nd)$; typically 10–20 iterations suffice.*

RL/ML Connection

The conjugate gradient approach is exactly the technique used in TRPO [Schulman et al., 2015], where the Fisher-vector product is computed by automatic differentiation through the KL divergence Hessian, avoiding explicit construction of the $d \times d$ Fisher matrix.

9 Performance Improvement Bound

The natural gradient enjoys a stronger performance improvement guarantee than the vanilla gradient, as established by Kakade [2002].

Theorem 9.1 (Policy Improvement under Natural Gradient)**.** *Let $\theta' = \theta + \alpha\, \tilde{\nabla}_\theta J(\theta)$ be the natural gradient update. For small α:*

$$J(\theta') - J(\theta) \geq \alpha\, \nabla_\theta J(\theta)^\top F(\theta)^{-1}\, \nabla_\theta J(\theta) - O(\alpha^2).$$

Since $\nabla_\theta J^\top F^{-1} \nabla_\theta J = \|F^{-1/2} \nabla_\theta J\|^2 \geq 0$ (with equality iff $\nabla_\theta J = 0$, assuming $F \succ 0$), the natural gradient guarantees performance improvement at every step for sufficiently small α.

Proof. By Taylor expansion: $J(\theta') = J(\theta) + \nabla_\theta J^\top (\theta' - \theta) + O(\|\theta' - \theta\|^2)$. Substituting $\theta' - \theta = \alpha\, F^{-1}\, \nabla_\theta J$ gives the result. □

10 Worked Example: Natural Gradient on a Bandit

Example 10.1 (Comparing Vanilla and Natural Gradients)**.** Consider a bandit with $\mathcal{A} = \{0, 1\}$, rewards $r(0) = 1$, $r(1) = 3$, and softmax policy $\pi_\theta(1) = \sigma(\theta) = p$. At $\theta = 0$ (so $p = 0.5$):

Vanilla gradient. From Lesson 55, Example 3.3: $\nabla_\theta J = 2p(1-p) = 0.5$.

Fisher information. For the scalar Bernoulli case, the score for $a = 1$ is $1 - p$ and for $a = 0$ is $-p$, so:

$$F(\theta) = p(1-p)^2 + (1-p)p^2 = p(1-p)[(1-p) + p] = p(1-p) = 0.25.$$

Natural gradient. $\tilde{\nabla}_\theta J = F^{-1} \nabla_\theta J = \frac{0.5}{0.25} = 2.0$.

The natural gradient is 4 times larger. At $\theta = 3$ ($p \approx 0.953$): $\nabla_\theta J \approx 0.090$, $F \approx 0.045$, $\tilde{\nabla}_\theta J = 2.0$. The natural gradient remains constant at 2.0 regardless of θ, while the vanilla gradient shrinks as the policy becomes deterministic. The natural gradient correctly identifies that the same improvement in distribution space requires larger parameter steps near determinism.

11 Exercises

Exercise 11.1 (Fisher Information for Bernoulli Policy)**.** $(*)$
For a Bernoulli policy $\pi_\theta(a = 1 \mid s) = \sigma(\theta^\top \phi(s))$ with sigmoid function σ, show that the Fisher information at state s is

$$F_s(\theta) = \sigma(\theta^\top \phi(s))\,(1 - \sigma(\theta^\top \phi(s)))\,\phi(s)\,\phi(s)^\top.$$

Verify that F_s is positive semi-definite with rank at most 1.

Exercise 11.2 (KL Divergence Hessian Verification)**.** $(*)$
For the softmax policy with $\mathcal{A} = \{0, 1\}$ and scalar parameter θ, explicitly compute $\mathrm{KL}(\pi_\theta \| \pi_{\theta'})$ as a function of θ and θ', then verify that $\frac{\partial^2}{\partial \theta'^2} \mathrm{KL}(\pi_\theta \| \pi_{\theta'})\big|_{\theta'=\theta} = F(\theta) = \sigma(\theta)(1 - \sigma(\theta))$.

Exercise 11.3 (Natural Gradient for Gaussian with Learnable Variance)**.** $(**)$
Consider a Gaussian policy with parameters $\theta = (\mu, \log \sigma) \in \mathbb{R}^2$ (scalar state and action): $\pi_\theta(a) = \mathcal{N}(a;\, \mu,\, \sigma^2)$.

(a) Compute the 2×2 Fisher information matrix $F(\theta)$.

(b) Show that F is diagonal, i.e., the mean and log-variance parameters are "orthogonal" in the Fisher metric.

(c) For the objective $J = \mathbb{E}_\pi[r(a)]$ with $r(a) = -a^2$, compute the vanilla and natural gradients at $\mu = 1$, $\sigma = 1$.

Exercise 11.4 (Compatible Features and Advantage Regression)**.** $(**)$
Let π_θ be a softmax policy with features $\phi(s, a) \in \mathbb{R}^d$. Consider the compatible function approximator $f_w(s, a) = w^\top(\phi(s, a) - \bar{\phi}_\theta(s))$.

(a) Show that f_w automatically satisfies $\mathbb{E}_{a \sim \pi_\theta}[f_w(s, a)] = 0$ for all s, confirming it approximates the advantage (not the Q-function).

(b) Write the normal equations for w^* and verify they give $w^* = F^{-1} \nabla_\theta J$.

Exercise 11.5 (Convergence Rate of Natural Policy Gradient)**.** $(***)$
Suppose the Fisher information matrix satisfies $\mu_{\min} I \preceq F(\theta) \preceq \mu_{\max} I$ for all θ, with $0 < \mu_{\min} \leq \mu_{\max}$. Show that the natural gradient satisfies the "gradient domination" bound:

$$\|\tilde{\nabla}_\theta J\|^2 \geq \frac{1}{\mu_{\max}^2} \|\nabla_\theta J\|^2,$$

and use this to argue that the natural gradient converges at least as fast as the vanilla gradient (in terms of the norm of the Euclidean gradient) up to a condition number factor $\kappa = \mu_{\max}/\mu_{\min}$.

Hint: Compare the per-step improvement $\nabla J^\top F^{-1} \nabla J$ to $\|\nabla J\|^2/\mu_{\max}$.

12 Summary

- The **vanilla policy gradient** is not covariant under reparameterization, leading to optimization sensitivity.
- The **Fisher information matrix** $F(\theta) = \mathbb{E}[\nabla \log \pi_\theta \, (\nabla \log \pi_\theta)^\top]$ defines a Riemannian metric on the policy manifold and equals the Hessian of the KL divergence between neighboring policies.
- The **natural gradient** $\tilde{\nabla}_\theta J = F(\theta)^{-1} \, \nabla_\theta J(\theta)$ is the steepest ascent direction under a KL divergence constraint and is covariant under smooth invertible reparameterizations.
- The natural gradient equals the **optimal compatible function approximation weight vector**, providing a practical alternative to explicit Fisher matrix inversion.
- For **Gaussian policies**, the natural gradient removes the $1/\sigma^2$ scaling; for **softmax policies**, the Fisher information is the feature covariance under the current policy.
- The natural gradient guarantees **monotonic improvement** (for small step sizes) and provides a foundation for trust-region methods.

13 Further Reading

- **Lesson 57** develops actor-critic architectures and Generalized Advantage Estimation (GAE), which combines with the natural gradient in the natural actor-critic algorithm.
- **Lesson 58** extends the natural gradient idea to trust-region methods, leading to TRPO [Schulman et al., 2015].
- ichi Amari [2016] provides a comprehensive treatment of information geometry, including the Fisher metric and connections to exponential families.
- Peters and Schaal [2008] develops the natural actor-critic with compatible features and TD-based critic updates.
- Kakade [2003, Chapter 2] treats the sample complexity of natural policy gradient methods.

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